MIT ZARAGOZA
International Logistics Program

MZ505 Inventory Systems Fall 2026

Assignment #5

Prof. Spyros Lekkakos

- **Individual Assignment**
- **Due: Nov 9th (Mon): 08.30 AM**
- **Please submit you work online via Canvas (only one .pdf file) and provide a hard copy before class.**

Problem 1 (Max Points 70)

Handi Inc. is a manufacturer of electronic devices in the United States. The company has a single manufacturing facility in San Jose, California. Handi distributes its products through five regional warehouses located in Los Angeles, Boston, Chicago, Dallas, and Atlanta. In the current distribution system, the US has five major markets, each of which is served by the single regional warehouse in their market. That is, currently, each customer is assigned to a single market and receives products from one regional warehouse. The warehouses receive products from the manufacturing facility. Typically, it takes two weeks to satisfy an order placed by any of the regional warehouses. Currently, Handi provides their customers with a service level of about 90 percent.

Recently, Handi has seen a significant increase in competition and intense pressure from its customers to reduce costs and improve service level. To achieve these goals, Handi would like to consider an alternative distribution strategy in which the five regional warehouses are replaced with a single, central warehouse that would process all customer orders. This warehouse should be one of the five existing warehouses. The company insists that whatever distribution strategy is used, it should attain an increased service level of about 97 percent. Please address the following questions:

(a) An analysis of customer demand reveals that the demands in the five regions are very similar; i.e., it is common that if weekly demand in one region is above average, so is the weekly demand in the other regions. How does this observation affect the attractiveness of the new system? Please **only** explain (**no correlation calculations on the data set needed**).

(b) Consider a typical product A. Table 1 provides historical data that includes weekly demand for this product for the last 12 weeks in each of the market areas. An order (placed by a warehouse to the facility) costs $5,500, and inventory holding costs are $1.25 per unit per week. Table 2 gives the cost of inbound and outbound transportation costs per unit in the current distribution system. Finally, Table 3 provides information about outbound transportation costs per unit from each of the existing regional warehouses to all the other market areas (assuming a particular regional warehouse becomes the centralized warehouse).

Table 1. Weekly Demand Historical Data

	Week											
	1	2	3	4	5	6	7	8	9	10	11	12
Atlanta	33	45	37	38	55	30	18	58	47	37	23	55
Boston	26	35	41	40	46	48	55	18	62	44	30	45
Chicago	44	34	22	55	48	72	62	28	27	95	35	45
Dallas	27	42	35	40	51	64	70	65	55	43	38	47
LA	32	43	54	40	46	74	40	35	45	38	48	56

Table 2. Transportation costs per unit product

Warehouse	**Inbound**	**Outbound**
Atlanta	12	13
Boston	11,5	13
Chicago	11	13
Dallas	9	13
LA	7	13

Table 3. Outbound transportation costs per unit in centralized system

Warehouse	Atlanta	Boston	Chicago	Dallas	LA
Atlanta	13	14	14	15	17
Boston	14	13	8	15	17
Chicago	14	8	13	15	16
Dallas	15	15	15	13	8
LA	17	17	16	8	13

Suppose you are to compare the two systems for Product A only, what is your recommendation? To answer this question, you should compare the total costs (i.e., ordering cost, inventory holding cost, inbound and outbound transportation costs) for the two strategies, assuming demands occur according to the historical data (you can assume here normally distributed independent demands and obtain the values for mean and standard deviation from Excel). In addition, you should determine which regional warehouse should be used as the centralized warehouse. (Assume **continuous review** system).

Problem 2 (Max Points 30)

You are in charge of designing a supply chain for furniture distribution. One of your products is a study desk. This desk comes in two colors: black and cherry. Weekly demand for each desk type is normal with mean 100 and standard deviation 65 (demands for the two colors are independent). The lead time from the assembly plant to the retail store is two weeks and you order inventory replenishments weekly. There is no finished goods inventory at the plant (desks are assembled to order for delivery to the store).

(a) What is the expected on-hand inventory of desks at the store (black and cherry together) if you maintain a 97% in-stock probability for each desk color?

You notice that only the top part of the desk is black or cherry; the remainder (base) is made of the standard gray metal. Hence, you suggest that the store stock black and cherry tops separately from gray bases and assemble them when demand occurs. The replenishment lead time for components is still two weeks. Furthermore, you still choose an order-up-to level for each top to generate a 97% in-stock probability.

(b) What is the expected on-hand inventory of black tops?

(c) How much less inventory of gray bases do you have on average at the store with the new in-store assembly scheme relative to the original system in which desks are delivered fully assembled?